

REVIEW

Are seed oils really bad for you?

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ABSTRACT

Ordinary culinary use of seed oils is unlikely to cause chronic disease, and replacing fats rich in saturated or trans fatty acids with unsaturated oils improves cardiovascular risk markers. Controlled diets consistently lower low-density lipoprotein cholesterol; one trial meta-analysis estimated a coronary-event risk ratio of 0.81, although later reviews judged the event evidence less certain. Two recovered trials reported no benefit or possible harm, but their old, intensive interventions and incomplete exposure records do not describe contemporary use. Human trials show no consistent inflammatory or glycaemic penalty, and prospective studies do not identify a broad cancer or mortality signal. The conclusion is conditional: oil composition, the displaced food, total energy, and heating history all matter. Fresh oil used at controlled temperatures is a different exposure from repeatedly heated frying oil, while a food containing seed oil is not healthy merely because its fatty acids are unsaturated.

01 Introduction

“Seed oil” is a culinary label, not a defined intervention. Conventional sunflower and corn oils are rich in linoleic acid, whereas high-oleic sunflower oil and canola oil provide different proportions of monounsaturated, polyunsaturated, and saturated fatty acids.^{1,2} Reviews that compare named oils therefore cannot assign one composition or one physiological action to the category.³ Processing and repeated heating add further compositional differences.¹

The strongest claims against seed oils combine several questions. One concerns replacement: what happens when an unsaturated oil displaces butter, palm oil, or carbohydrate? A second concerns food context: can the oil ingredient explain the health associations of an ultra-processed diet? A third concerns thermal history: does repeated heating generate products that are absent from fresh oil? Controlled oil interventions and food-substitution analyses separate some of these exposures and show why the ingredient label alone is an unreliable proxy for processing.^{4,5,6}

The central question is therefore not whether a botanical category is intrinsically good or bad. It is whether a specified oil, used in a specified way, improves or worsens health relative to the food or fat it replaces. The evidence becomes more consistent once that comparison is kept visible.

02 The label “seed oils” hides biologically different exposures

Oils pressed from seeds can differ more from one another than two versions of the same crop. Breeding for high oleic acid changes the balance between monounsaturated and polyunsaturated fatty acids, while refining, antioxidants, storage, and cooking alter the compounds that reach a meal.^{2,1} These differences are large enough to change lipid

responses and oxidation under heat. A claim about “seed oils” that does not specify the oil and its preparation leaves its causal exposure undefined.

That imprecision also obscures dose. Adding oil increases energy intake unless another food is removed, but most controlled trials maintain energy while exchanging one fat source for another.^{7,8} Observational substitution models make the same contrast mathematically. Neither design answers what happens when oil is simply poured onto an otherwise unchanged diet. Figure synthesis therefore has to begin with the alternative food, not with the seed source.

The three exposure paths and the evidence each can support are separated in [Figure 1](#).

03 Replacement determines what the body experiences

Fat substitution is a zero-sum experiment at a fixed energy intake. When polyunsaturated fat rises, saturated fat, trans fat, carbohydrate, or another unsaturated fat must fall. Prospective cohorts that model a 2%–5% energy exchange and controlled-feeding trials both show that the chosen comparator changes the estimated effect.^{8,9} A smaller difference between two unsaturated oils does not contradict a larger difference between an unsaturated oil and butter.

Carbohydrate is not a single comparator either. Replacing saturated fat with refined starch can produce a different lipid response from replacing it with intact carbohydrate-rich foods, while exchanging saturated fat for polyunsaturated fat directly changes both sides of the fatty-acid balance.⁷ Quantitative reviews of controlled diets therefore express effects as nutrient exchanges rather than isolated additions. This design choice is causal bookkeeping: without the displaced source, even a precisely measured oil intake does not identify the intervention.

The recurring comparisons align as follows.

INCREASE	REPLACE	MAIN RESPONSE	EVIDENCE
PUFA oil	Saturated fat	LDL falls	10
Corn oil	Saturated fat	Lipids improve	11
Canola oil	Saturated fat	Vascular test null	12
Plant foods	Animal foods	Risk estimates improve	6
Oil calories	Nothing specified	Causal effect unclear	7

Food replacement also prevents a common reversal of inference. A cohort association between fried food and disease cannot establish that its oil ingredient caused the association, because the food carries refined starch, salt, energy density, and a cooking history. Conversely, a favorable lipid response to purified oil does not establish that every pack-

aged food containing it is favorable. The ingredient participates in both settings, but it does not define either one. This dependence on the displaced food is the first result that travels across the literature. It determines how the lipid trials should be read and why studies with apparently similar oil exposures can point in different directions.

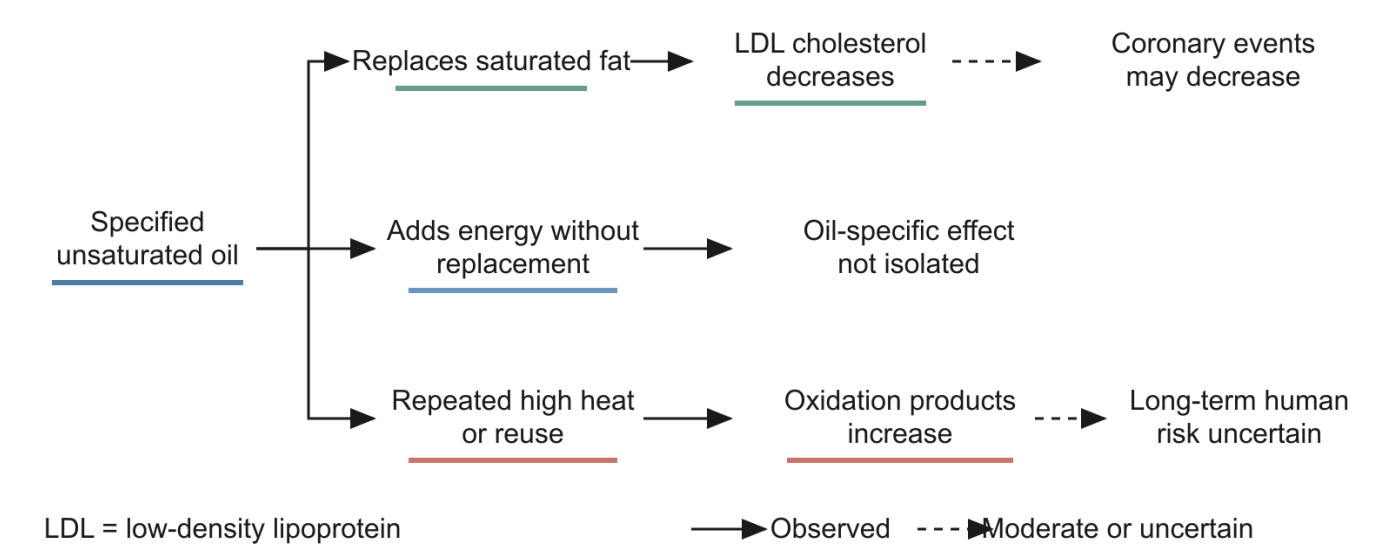


Figure 1. Health effects depend on the comparison and the heating history. The strongest human evidence follows an energy-matched replacement of saturated fat with unsaturated oil; simply adding calories is a different, poorly isolated contrast. Repeated heating creates a separate chemical exposure whose long-term human consequences remain uncertain. ^{7,6,13,1}

04 Unsaturated-oil substitutions reliably lower LDL-related lipoproteins

Controlled diets consistently show that replacing saturated fat with linoleic-acid-rich or other unsaturated oils lowers low-density lipoprotein (LDL) cholesterol and related atherogenic lipoproteins. ^{10,5} Trials of rapeseed, corn, soybean, sunflower, walnuts, and mixed unsaturated-fat diets converge on that direction even when their non-lipid outcomes differ. ^{14,15} Low-saturated-fat vegetable oils also produce more favorable LDL concentrations than palm oil in controlled comparisons. ¹⁶

The magnitude depends on the contrast. In a controlled walnut study, the walnut diet lowered LDL cholesterol by 6.9 mg/dL more than an oleic-acid comparison diet, while particle subclasses, cholesterol efflux, and several related markers did not differ. ¹⁷ A canola crossover trial likewise

improved the fat profile of the diet but found no difference in flow-mediated dilation (P=0.81). ¹² The artery-wall test and the cholesterol measure answered different questions. LDL is an established causal contributor to atherosclerotic disease, so its reduction supports cardiovascular benefit. ¹⁸ Yet an oil-specific trial must still run long enough and retain enough dietary separation to show fewer clinical events. The marker result is robust; the size of the event benefit is the next source of uncertainty.

05 Hard-outcome trials rule out a large hazard but leave modest benefit uncertain

Systematic reviews of randomized trials do not show that increasing omega-6 or total polyunsaturated fat causes a large rise in cardiovascular events or deaths. Their estimates usually allow little or no effect and, in some analyses, a modest event reduction; trial quality, incomplete reporting,

and heterogeneous oil mixtures lower confidence.^{19,20,21} This is reassuring evidence against broad toxicity, but it is not a precise estimate for a named modern oil.

Clinical endpoints also differ in the amount of evidence they require. A cholesterol concentration can change within weeks in a feeding study, whereas coronary events accumulate slowly and require large samples with sustained adherence. Several older trials supplied enough lipid measurements to establish biological separation but too few well-ascertained events to estimate mortality precisely.^{19,21} The event literature is therefore smaller and noisier for a struc-

tural reason, not because the lipid result disappears at longer follow-up.

Dariusz Mozaffarian and colleagues pooled eight trials with 13,614 participants and 1,042 coronary events. Increasing polyunsaturated fat in place of saturated fat yielded a coronary-event risk ratio of 0.81 (95% confidence interval, 0.70–0.95), with about 10% lower risk for each 5% of energy exchanged.²² Later syntheses were more cautious because they included different trials and judged the older methods more severely.^{23,19} The disagreement concerns the size and certainty of benefit more than evidence for a major class-wide harm.

EVIDENCE SET	PARTICIPANTS	CONTRAST	CLINICAL RESULT
Eight-trial pool	13,614	PUFA for saturated fat	Coronary RR 0.81 (0.70–0.95) ²²
Minnesota	9,423	Corn oil diet	Mortality not reduced ²⁴
Sydney	458	Safflower oil diet	Mortality HR 1.62 (1.00–2.64) ²⁵

These trials were conducted decades ago.^{25,24} Their designs and background diets narrow how directly they answer present-day oil choices, but they do not erase the randomized evidence. The two adverse or null reanalyses deserve separate treatment because their designs explain much of the dispute.

06 Two recovered trials explain much of the remaining cardiovascular dispute

The Sydney Diet Heart Study provides the clearest adverse result. In 458 men with previous coronary events, a safflower-oil intervention was associated with all-cause mortality of 17.6% versus 11.8% (hazard ratio 1.62, 95% confidence interval 1.00–2.64), with similar elevations for cardiovascular and coronary mortality.²⁵ The trial changed several dietary components, delivered linoleic acid near 15% of energy, and relied on incompletely recovered records. Its result is real within that experiment; its exposure is not ordinary household use.

Christopher E Ramsden and colleagues recovered data from the Minnesota Coronary Experiment, which randomized 9,423 institutionalized participants. The intervention lowered serum cholesterol by 13.8%, compared with 1.0% in controls, yet it did not reduce mortality; the pooled coronary-mortality hazard ratio was 1.13 (95% confidence interval 0.83–1.54).²⁴ Only 2,355 participants remained on the assigned diet for at least one year, and missing records limited endpoint reconstruction.

Trial age matters because the interventions, background diets, follow-up, and outcome ascertainment differ from those of a modern efficacy trial. Those limits are documented features of the recovered studies, rather than assumptions about how contemporary populations would have responded.^{25,24} Reviews that weigh all randomized evidence therefore treat Sydney as an adverse outlier and

Minnesota as an important null result, while retaining uncertainty about clinical benefit.^{18,20}

The old trials break a simple chain from lower LDL to guaranteed event reduction. They do not reverse the wider evidence into proof of harm.

07 Prospective intake and biomarker cohorts do not reproduce a broad hazard

Prospective dietary cohorts generally associate higher linoleic-acid intake with neutral or lower coronary and mortality risk.^{26,27} Meta-analyses of intake, coronary disease, and total mortality point in that direction, although food questionnaires, modeled substitutions, and residual confounding prevent causal language.²⁸ Individual cohorts can be null, and estimates vary with the chosen replacement nutrient.^{29,30}

Biomarkers reduce recall error but measure more than diet. Circulating and tissue linoleic acid also reflect absorption, metabolism, genetics, and the interval since eating. Even with those limits, pooled biomarker analyses and large cohorts repeatedly find neutral-to-inverse associations with cardiovascular disease and mortality.^{31,32,33} The Framingham Offspring analysis followed 2,500 adults for up to 9.5 years and recorded 245 cardiovascular events and 350 deaths; erythrocyte omega-6 fatty acids showed little association with either outcome.³⁴ More recent plasma analyses likewise do not reveal a broad positive association.^{35,36}

Diet questionnaires and biomarkers also fail in different ways. Self-reported intake can misclassify oils hidden in mixed foods, while a biomarker may be influenced by metabolism even when intake is stable. Agreement across those measurement systems reduces the chance that one shared measurement error created the result, although healthy-user behavior and residual confounding can still affect

both.^{31,26} The convergence is informative because the biases are not identical.

Stroke provides an outcome-specific example. Adipose linoleic acid in a Danish case-cohort study was associated with less ischemic stroke (highest versus lowest quartile hazard ratio 0.78, 95% confidence interval 0.65–0.93).³⁷ Borges and colleagues pooled UK data and found no independent coronary association but a lower odds of stroke per standard-deviation increase in circulating linoleic acid (odds ratio 0.82, 95% confidence interval 0.75–0.90).³⁸ A separate biomarker meta-analysis supports the same direction.³⁹

A ratio can obscure this pattern. In UK Biobank, the highest omega-6-to-omega-3 ratio quintile was associated with higher all-cause mortality (hazard ratio 1.26, 95% confidence interval 1.15–1.38), yet omega-6 and omega-3 concentrations were each independently associated with lower mortality.⁴⁰ The ratio cannot reveal whether its value rose because omega-6 increased, omega-3 fell, or both changed. Examining each fatty acid preserves the information the ratio discards.

Observational studies cannot supply the randomization missing from old outcome trials. Their consistent absence of a positive signal is nevertheless difficult to reconcile with a large, common toxic effect.

08 Human intervention studies do not show an inflammatory penalty

The simplest inflammatory theory proposes that dietary linoleic acid increases arachidonic acid and then pro-inflammatory mediators. Human intervention data do not support that pathway as an automatic net response. Systematic reviews of randomized trials find no consistent increase in C-reactive protein, interleukin-6, tumor-necrosis factor, or related circulating markers when linoleic acid rises.^{41,42,43}

Human fatty-acid metabolism regulates conversion rather than passing dietary linoleic acid through an unrestricted one-way channel.^{44,45} Reviews of arachidonic acid, eicosanoids, and specialized lipid mediators show that both pro-resolving and pro-inflammatory pathways exist; genotype, omega-3 intake, and disease state can shift them. Biochemical plausibility therefore survives, but the predicted systemic inflammatory increase does not appear in conventional human biomarkers.

Standard blood markers cannot describe every tissue signal, and most trials are short. Those boundaries justify continued mechanistic work. They do not justify substituting a pathway diagram for the intervention result.

09 Metabolic outcomes are neutral to favorable when unsaturated fat replaces saturated fat

Controlled-feeding studies do not show worsening glucose control when plant-derived polyunsaturated fat replaces saturated fat or carbohydrate.⁴⁶ Across 13 trials, fasting glucose changed by -0.01 mmol/L (95% confidence inter-

val -0.06 to 0.03), fasting insulin by -2.6 pmol/L (-4.9 to -0.2), and the homeostatic model assessment of insulin resistance by -0.12 units (-0.23 to -0.01).⁴⁶ A broader randomized review found the omega-6-specific diabetes evidence sparse but no major glycaemic harm.⁴⁷

Liver fat moved in the same direction in controlled experiments. Trials in abdominal obesity and overfeeding found less liver-fat accumulation when polyunsaturated fat replaced saturated fat.^{48,49} In a 12-month trial of 150 adults with prediabetes or type 2 diabetes, a high-polyunsaturated-fat dietary program reduced liver fat by 1.46 percentage points (95% confidence interval -2.42 to -0.51) relative to usual care.⁵⁰ The intervention changed whole diets, so it cannot be assigned to one bottle of oil.

Prospective studies associate higher linoleic-acid intake or biomarker patterns with neutral or lower type 2 diabetes risk.^{51,52} Among 11,264 adults with diabetes, modeled replacement of 2% of energy from saturated fat with linoleic acid was associated with cardiovascular-mortality hazard ratio 0.85 (95% confidence interval 0.73–0.99).⁵³ Diabetes intervention reviews remain heterogeneous because their oils, comparators, and background treatments differ.⁵⁴

The metabolic evidence is strongest for ruling out the proposed glucose penalty. Small improvements in insulin or liver fat remain contingent on what the unsaturated fat displaced.

10 Cancer data exclude a large general signal more clearly than a small site-specific effect

Cancer outcomes require longer follow-up and greater precision than most dietary trials provide. A meta-analysis of 70 prospective reports found essentially null risk for dietary omega-6 intake (relative risk 1.02, 95% confidence interval 0.99–1.05) and an inverse association for blood omega-6 biomarkers (relative risk 0.92, 0.86–0.98).⁵⁵ Cross-outcome reviews likewise found no consistent general cancer signal.¹⁸

Randomized estimates are much less precise. Across 4,272 participants in omega-6-specific trials, the relative risk for any cancer was 1.21, with a confidence interval from 0.96 to 1.53.⁵⁶ That interval includes little difference and clinically relevant harm. The trials therefore cannot prove equivalence, even though the observational literature does not show the broad increase predicted by a strong carcinogenic effect.

Site-specific meta-analyses for breast and prostate cancer also fail to identify a consistent linoleic-acid association.^{57,58} Tumor subtypes, long latency, and exposure measurement leave room for small effects. The available data answer the sweeping claim more firmly than every possible site-specific question.

11 Heat exposure changes the question because oxidation rises with temperature and reuse

Polyunsaturated oils oxidize more readily than monounsaturated-rich oils during prolonged high-temperature cooking.^{13,59} Laboratory studies detect aldehydes and other degradation products as temperature, time, and reuse increase.⁶⁰ Antioxidants, oil replenishment, and high-oleic formulations modify that response, so fatty-acid class alone does not determine the final mixture.¹

Saiuj Bhat and colleagues pooled 33 experimental heating studies. Below 200°C, oils formed little trans fat; from 200°C to 240°C, total trans fat rose by 0.38 percentage points for every 10°C increase (95% confidence interval 0.20–0.55).¹³ Evidence above 240°C was sparse. The temperature gradient and the separate role of repeated use are shown in Figure 2.

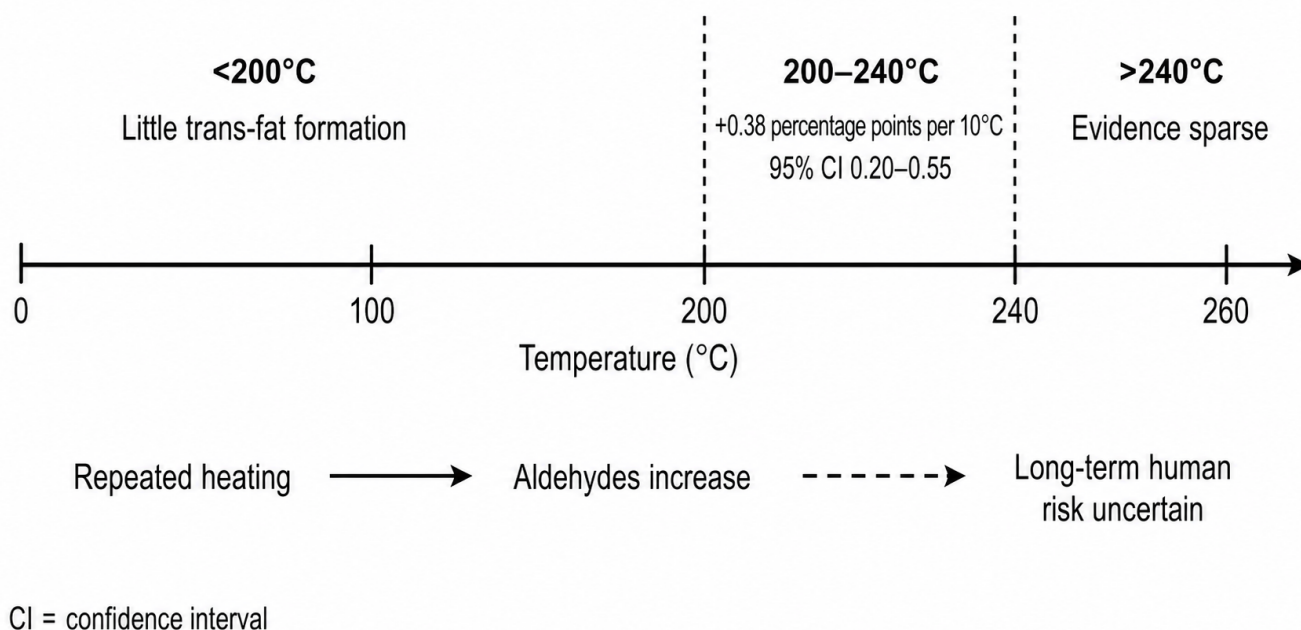


Figure 2. Oxidation and trans-fat formation rise with thermal stress. The horizontal scale reports the observed meta-analytic temperature gradient, while the separate reuse path represents additional degradation that temperature alone does not capture. Chemistry is well documented; long-term human disease risk at specific household exposures is not.^{13,59,61,62}

Repeatedly heated oil is consequently a distinct exposure. Reviews, occupational biomarker studies, and short meal experiments support minimizing reuse, although no long-term randomized human trial quantifies the resulting disease reduction.^{61,62,63,64} One frying experiment found roughly 10–25 parts per million of each monitored aldehyde class in restaurant fries, but the concentration varied by oil and protocol.⁵⁹

Normal home cooking cannot be assigned one universal temperature, but fresh oil, limited duration, and avoidance of smoking define a different regime from commercial or repeated frying. The chemistry supports a precaution against thermal abuse without converting every sauté into a toxic exposure.

12 Food matrix and oil formulation limit ingredient-level conclusions

Whole foods can produce effects that isolated fatty acids do not. In a controlled crossover study, whole walnuts reduced central diastolic pressure by 1.78 mm Hg while an oleic-acid replacement changed it by 0.15 mm Hg ($P=0.04$); systolic pressure and arterial stiffness were unchanged.⁶⁵

Fiber, minerals, and other walnut constituents prevent assignment of the pressure result to fatty-acid composition alone. That is precisely why food matrices matter.

Named oils also differ in short feeding trials. Comparisons among soybean, olive, camellia, canola, corn, and high-oleic oils show some lipid differences but frequent null results for vascular or metabolic endpoints.^{4,12,66} Reviews of conventional and high-oleic oils do not support a stable rank that applies across every outcome.² The practical comparison remains outcome- and comparator-specific.

Processing level remains a property of the food, not of one fatty acid. Seed oils appear in controlled diets that improve risk markers and in packaged foods that combine refined starch, salt, and high energy density.^{5,6} A study that measures only the food pattern cannot distribute its association among those components, while an oil trial deliberately removes much of that context. Both designs are useful when they are kept within their own questions.

Lipidomics provides a more detailed surrogate view. In linked cohort and randomized analyses, 69 plasma lipid features predicted at least one outcome, and 17 of 19 diet-

responsive risk-associated lipids shifted favorably when unsaturated fat replaced saturated fat.^{67,68} Platform limits, secondary analyses, and the absence of clinical endpoints keep this evidence below the event trials. The molecular data reinforce the substitution pattern without settling the event question.

Current reviews that integrate trials and biomarker consortia reach a similar boundary: ordinary linoleic-acid-rich oils appear beneficial or neutral in place of saturated fat, while oil-specific endpoint trials remain scarce.^{69,70,28} Their convergence matters in clinical practice because it spans different measurements. It does not make the food label sufficient.

13 Conclusion

The literature repeatedly separates according to the comparator and the exposure definition. LDL trials, prospective cohorts, inflammatory-marker studies, metabolic experiments, and broad cancer analyses do not support treating ordinary seed oils as a class-wide toxin. The clinical-event trials are older and less decisive: one pooled analysis favors modest coronary benefit, Minnesota is null, and Sydney is adverse. Together they leave the size of benefit uncertain while making a large general hazard unlikely.

This pattern yields a practical conclusion with clear limits. Replacing butter, animal fat, or another saturated-fat source with a mostly unsaturated oil is better aligned with current evidence than reversing that substitution.^{6,16} Total calories and the food carrying the oil still matter. Repeatedly overheating or reusing oil should be minimized because the chemical exposure changes as heat accumulates.^{13,1}

A modern long-duration randomized trial could narrow the remaining cardiovascular uncertainty only if it specifies the oil, keeps energy and food processing comparable, verifies sustained separation between diets, and measures clinical outcomes. Human studies that repeatedly measure household and commercial frying products could define the thermal boundary more precisely. Until then, the evidence supports choosing mostly unsaturated oils for appropriate substitutions, controlling heat, and judging the entire food.

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